

Software Quality Engineering

Test Report

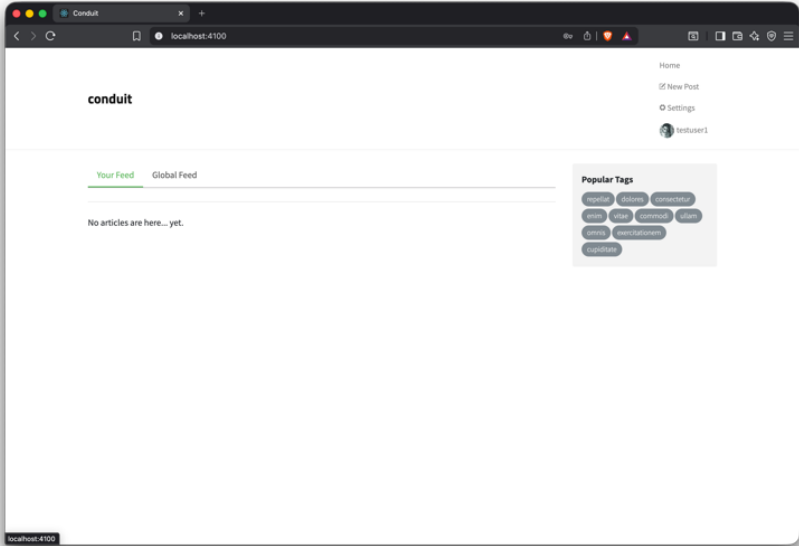
Project Name	Conduit RealWorld Blog App
Group Number	Group 3
Project Path	Path A — The Integrator
Tech Stack	React.js, Node.js, Express.js, MongoDB
Submission Date	10/6/24

Member 1	<i>Gnan Antopou</i>
Member 2	<i>Chron Rachana</i>
Member 3	<i>Than Soryapongs</i>
Member 4	<i>Nai Panhavorn</i>
Member 5	<i>Hor Bunnareth</i>

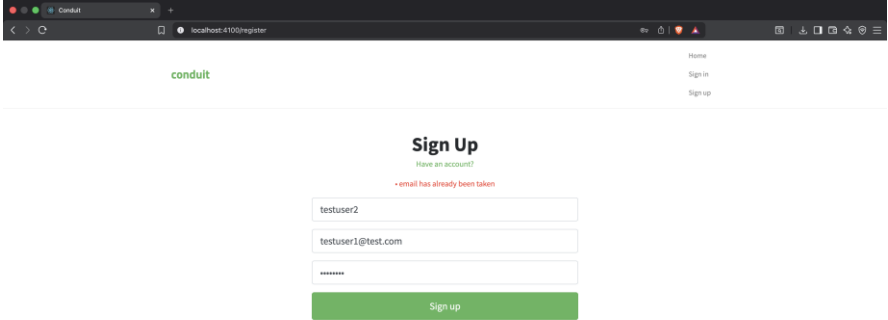
Phase 1A — Black-Box Test Cases

Each member tests their assigned cases, fills in the Actual Result, Status, and pastes their screenshot.

TC-001 — Register with valid data

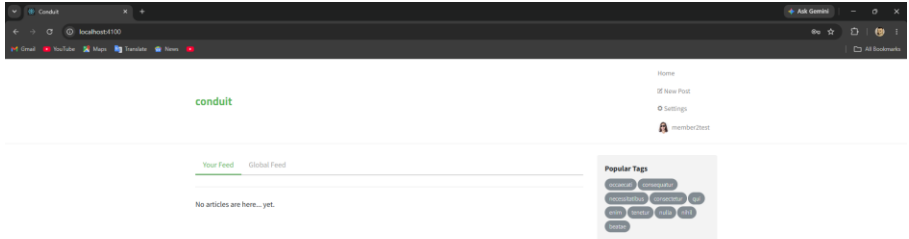
Test Case ID	TC-001
Test Title	Register with valid data
Tester Name	Gnan Antopou
Technique	Equivalence Partitioning
Pre-conditions	User is on the register page at http://localhost:4100/register
Test Steps	1. http://localhost:4100/register 2. Enter Username: testuser1 3. Enter Email: testuser1@test.com 4. Enter Password: Test1234 5. Click Sign Up
Test Data	Username: testuser1 Email: testuser1@test.com Password: Test1234
Expected Result	Account is created. User is redirected to home feed. Username appears in the navbar.
Actual Result	Account is created. User is redirected to home feed. Username appears in the navbar.
Status	Pass
Post-conditions	A new account exists in the system. User is logged in and on the home feed.
Screenshot	

TC-002 — Register with duplicate email

Test Case ID	TC-002
Test Title	Register with duplicate email
Tester Name	Gnan Antopou
Technique	Equivalence Partitioning — Sad Path
Pre-conditions	An account with email testuser1@test.com already exists in the system
Test Steps	1. http://localhost:4100/register 2. Enter Username: testuser2 3. Enter Email: testuser1@test.com 4. Enter Password: Test1234 5. Click Sign Up
Test Data	Username: testuser2 Email: testuser1@test.com (already used) Password: Test1234
Expected Result	An error message appears saying the email is already taken. Account is not created.
Actual Result	<i>An error message appeared saying email is already taken.</i>
Status	<i>Fail</i>
Post-conditions	<i>No new account was created. User remains on the register page.</i>
Screenshot	

TC-003 — Login with correct credentials

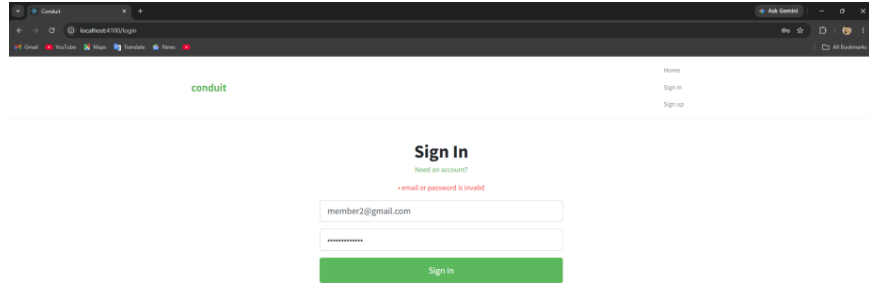
Test Case ID	TC-003
Test Title	Login with correct credentials
Tester Name	Chron Rachana
Technique	Equivalence Partitioning — Happy Path
Pre-conditions	A registered account exists. User is on the login page at localhost:4100/login

Test Steps	1. http://localhost:4100/login 2. Enter Email: member2@test.com 3. Enter Password: Test1234 4. Click Sign In
Test Data	Email: member2@test.com Password: Test1234
Expected Result	User is logged in. Home feed is displayed. Username shows in the top navbar.
Actual Result	User profile information is displayed.
Status	Pass
Post-conditions	User has access to authorized features
Screenshot	

TC-004 — Login with wrong password

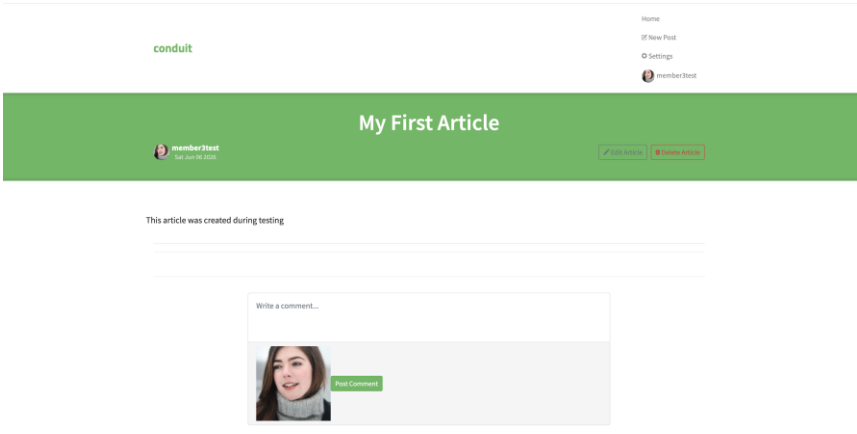
Test Case ID	TC-004
Test Title	Login with wrong password
Tester Name	Chron Rachana
Technique	Equivalence Partitioning — Sad Path
Pre-conditions	A registered account exists. User is on the login page
Test Steps	1. Go to localhost:4100/login 2. Enter Email: member2@test.com 3. Enter Password: wrongpassword 4. Click Sign In
Test Data	Email: testuser1@test.com Password: wrongpassword
Expected Result	An error message appears. User stays on the login page and is not logged in.
Actual Result	Error message " email or password is invalid " is displayed.
Status	Fail
Post-conditions	User cannot access protected pages.

Screenshot



TC-005 — Create article with all fields filled

Test Case ID	TC-005
Test Title	Create article with all fields filled
Tester Name	Than Soryapongs
Technique	Equivalence Partitioning — Happy Path
Pre-conditions	User is logged in and on the new article page at localhost:4100/#/editor
Test Steps	1. Login to the app 2. Click New Article in the navbar 3. Fill Title: My First Article 4. Fill Description: This is a test article 5. Fill Body: Article body content here 6. Fill Tags: testing 7. Click Publish Article
Test Data	Title: My First Article Description: This is a test article Tag: testing
Expected Result	Article is published. User is redirected to the article page. Article appears on their profile.
Actual Result	The same as Expected Result
Status	Pass
Post-conditions	New article exists in the database and is visible on the profile page.

Screenshot	 Did the article appear? Yes
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TC-006 — Create article with missing title

Test Case ID	TC-006
Test Title	Create article with missing title
Tester Name	Than Soryapongs
Technique	Boundary Value Analysis — Sad Path
Pre-conditions	User is logged in and on the new article page
Test Steps Test Data Expected Result	1. Login to the app 2. Click New Article 3. Leave Title field empty 4. Fill Description: No title test 5. Fill Body: Testing missing title 6. Click Publish Article Title: (empty) Description: No title test Body: Testing missing title An error message appears. Article is not published. User stays on the editor page.
Actual Result	<i>As Expected</i>
Status	Failed
Post-conditions	No article was created. Editor page remains open.

Screenshot

conduit

[Home](#)
[New Post](#)
[Settings](#)
member1234

Article Title

No title test

Testing missing title

Enter tags

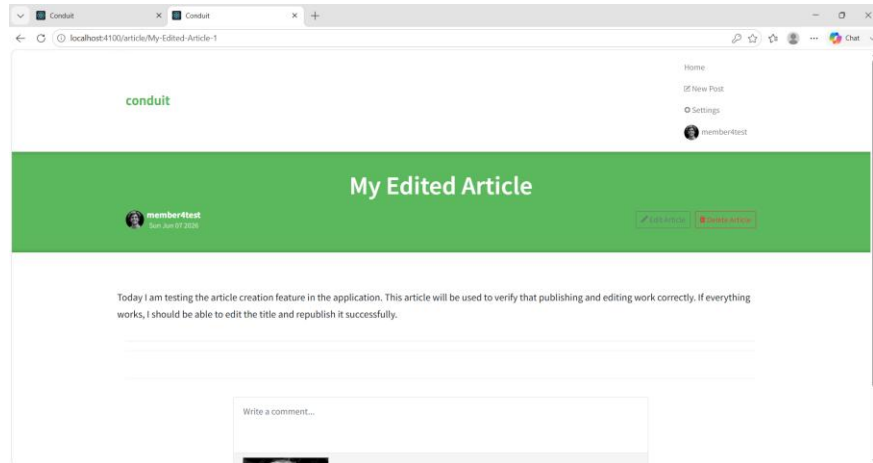
Publish Article

Did an error appear? Yes

TC-007 — Edit own article

Test Case ID	TC-007
Test Title	Edit own article
Tester Name	Nai Panhavorn
Technique	Equivalence Partitioning — Happy Path
Pre-conditions	User is logged in and has at least one published article
Test Steps	1. Login to the app 2. Go to your profile and open your article 3. Click Edit Article 4. Change title to: My Edited Article 5. Click Publish Article
Test Data	New Title: My Edited Article
Expected Result	Article is updated with the new title. Changes appear immediately on the article page.
Actual Result	The article title was successfully updated to “My Edited Article” and the changes were reflected immediately after publishing.
Status	Pass
Post-conditions	The updated article remains published with the new title saved in the system.

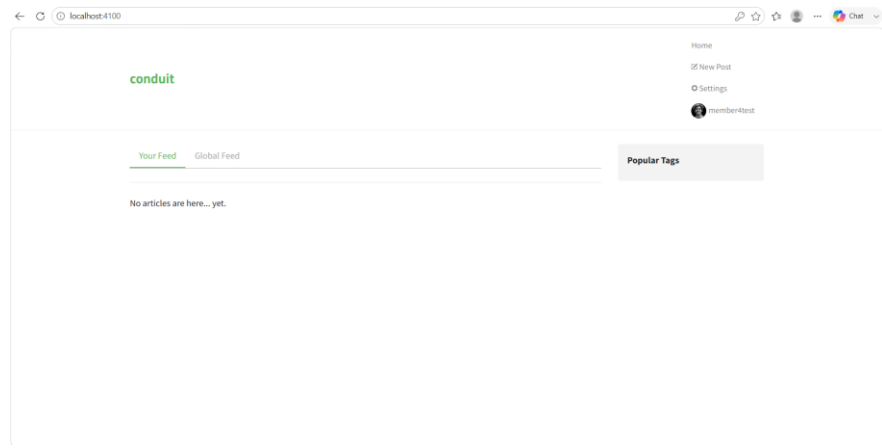
Screenshot



TC-008 — Delete own article

Test Case ID	TC-008
Test Title	Delete own article
Tester Name	Nai Panhavorn
Technique	Equivalence Partitioning — Happy Path
Pre-conditions	User is logged in and has at least one published article
Test Steps	1. Login to the app 2. Go to your profile and open your article 3. Click Delete Article
Test Data	Any article created by the logged-in user
Expected Result	Article is deleted. User is redirected. The article no longer appears on the profile page.
Actual Result	<i>The article was successfully deleted and removed from the profile page. The user was redirected after deletion.</i>
Status	Pass
Post-conditions	<i>The selected article is permanently removed from the user's article list.</i>

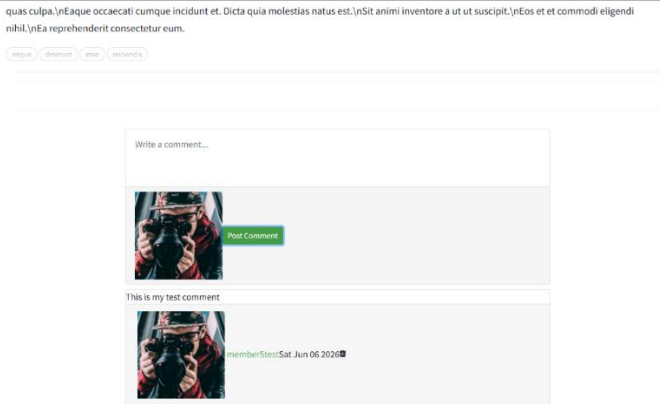
Screenshot



TC-009 — Follow another user

Test Case ID	TC-009
Test Title	Follow another user
Tester Name	Hor Bunnareth
Technique	Equivalence Partitioning — Happy Path
Pre-conditions	User is logged in. At least one other user account exists in the system
Test Steps	1. Login to the app 2. Click on any article on the home page 3. Click on the author name 4. Click the Follow button on their profile
Test Data	Any existing author profile in the app
Expected Result	Follow button changes to Unfollow. The author is added to the following list.
Actual Result	The Follow button changed to "Unfollow Jakub Kamau" after clicking the Follow button.
Status	Pass
Post-conditions	The logged-in user member5test is following Jakub Kamau.
Screenshot	

TC-010 — Add comment to an article

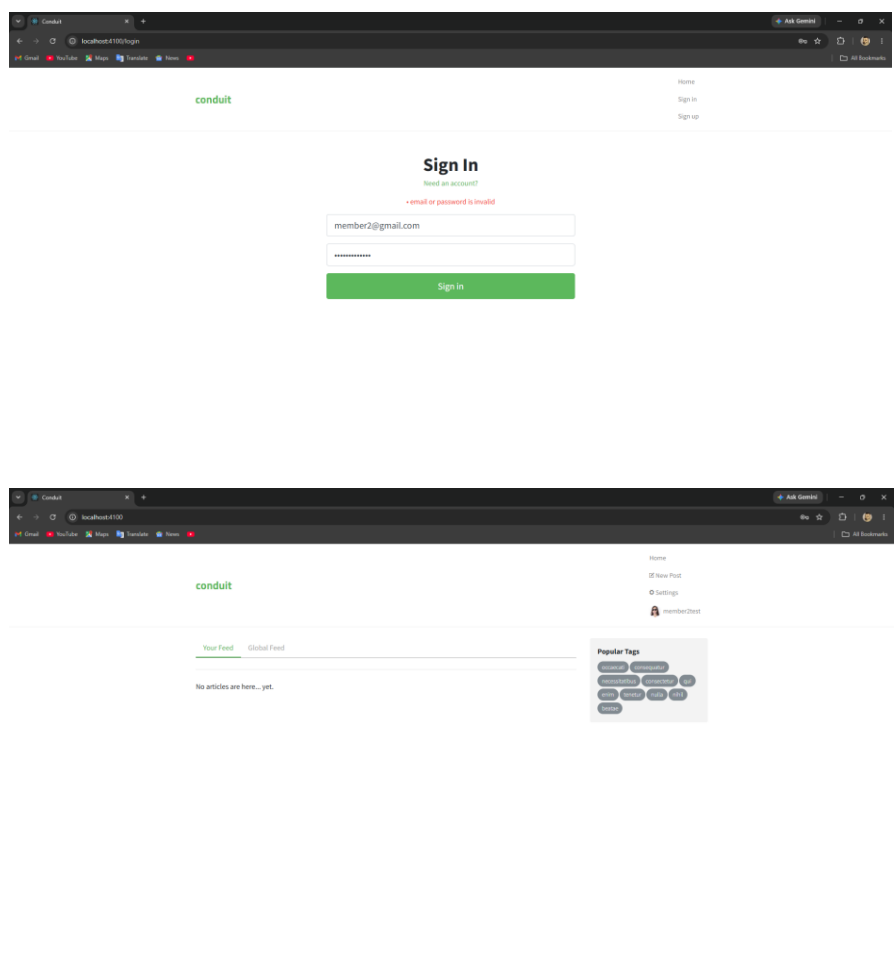
Test Case ID	TC-010
Test Title	Add comment to an article
Tester Name	<i>Hor Bunnareth</i>
Technique	Equivalence Partitioning — Happy Path
Pre-conditions	User is logged in. At least one article exists in the system
Test Steps	1. Login to the app 2. Click on any article on the home page 3. Scroll to the bottom 4. Type in the comment box: This is my test comment 5. Click Post Comment
Test Data	Comment text: This is my test comment
Expected Result	Comment appears below the article with the logged-in user's username and timestamp.
Actual Result	The comment "This is my test comment" appeared under the article with the logged-in username member5test and timestamp.
Status	Pass
Post-conditions	The new comment remains visible under the selected article.
Screenshot	Insert TC-010 screenshot showing the posted comment: "This is my test comment".
	 The screenshot displays a web interface for adding a comment. At the top, there is a text area with placeholder text: "quas culpa.\nEaque occaecati cumque incidunt et. Dicta quia molestias natus est.\nSit animi inventore a ut ut suscipit.\nEos et et commodi eligendi nihil.\nEa reprehenderit consectetur eum." Below the text area are four buttons: "unique", "denunciate", "esse", and "voluptas". A "Write a comment..." label is positioned above a text input field. To the right of the input field is a green "Post Comment" button. Below the input field, the text "This is my test comment" is visible. At the bottom, there is a user profile picture and the text "member5test Sat Jun 06 2020".

Phase 1B — Investigative Testing (Charter Logs)

Each charter is a free 20-30 minute exploration session. Document everything you tried and found.

CH-001 — Chron Rachana

Charter ID	CH-001
Tester Name	Chron Rachana
Mission	Explore login and registration to find broken validation or security issues
Areas to Explore	Register page, login page, form validation, error messages, special characters in fields, empty fields
Start Time	5:30 PM
End Time	7:00 PM
Duration	1:30 minutes
What I Tried	1. Opened the login page. 2. Entered a valid email address. 3. Entered the correct password. 4. Clicked the "Sign In" button. 5. Verified that the system redirected to the Home page.
Findings / Bugs	User was able to log in successfully without any issues.
Observations	The user session was created successfully. Navigation menu changed to show authenticated user options.



CH-002 — Than Soryapongs	
Charter ID	CH-002
Tester Name	Than Soryapongs
Mission	Hunt for bugs in article creation and editing — look for missing validation and data issues
Areas to Explore	New article editor, publish flow, edit article, tag system, markdown rendering, very long text inputs
Start Time	1:40 PM
End Time	2:04 PM
Duration	24 minutes
What I Tried	- Paste 10,000 characters into the title - Add multiple tags at once - Publish, then immediately edit and publish again
Findings / Bugs	- Tag is not storing into DB after published - Title only exceeded 100 characters error but not display error message to interface

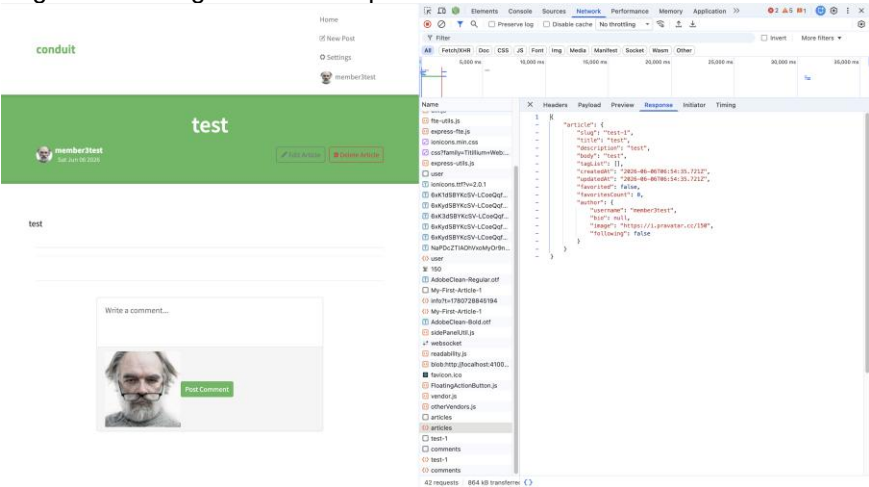
Charter ID	CH-002
Tester Name	Than Soryapongs
Mission	Hunt for bugs in article creation and editing — look for missing validation and data issues
Areas to Explore	New article editor, publish flow, edit article, tag system, markdown rendering, very long text inputs
Start Time	1:40 PM
End Time	2:04 PM
Duration	24 minutes
What I Tried	- Paste 10,000 characters into the title - Add multiple tags at once - Publish, then immediately edit and publish again
Findings / Bugs	- Tag is not storing into DB after published - Title only exceeded 100 characters error but not display error message to interface

Observations

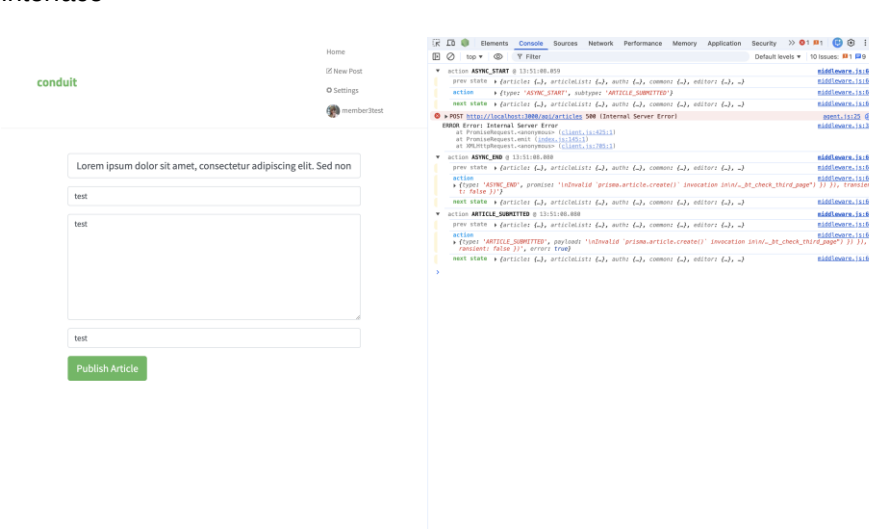
Screenshots

Inspect error in Browser DevTools

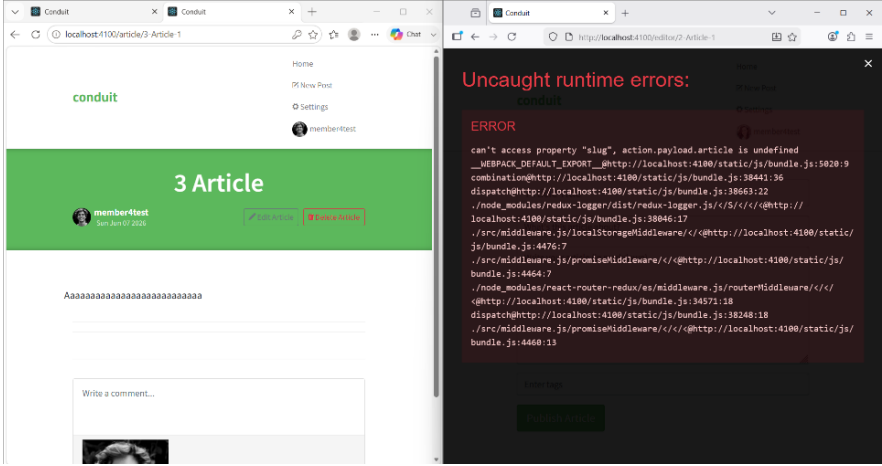
Tag is not storing into DB after published



Title only exceeded 100 characters error but not display error message to interface



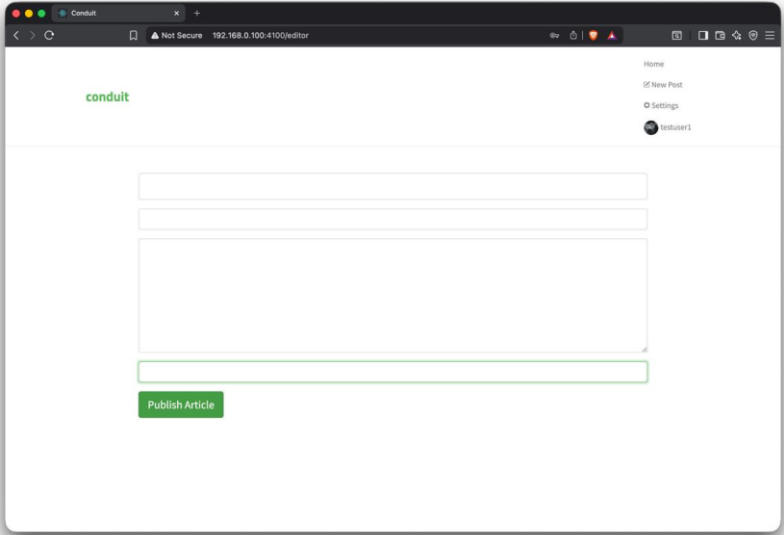
CH-003 — Nai Panhavorn

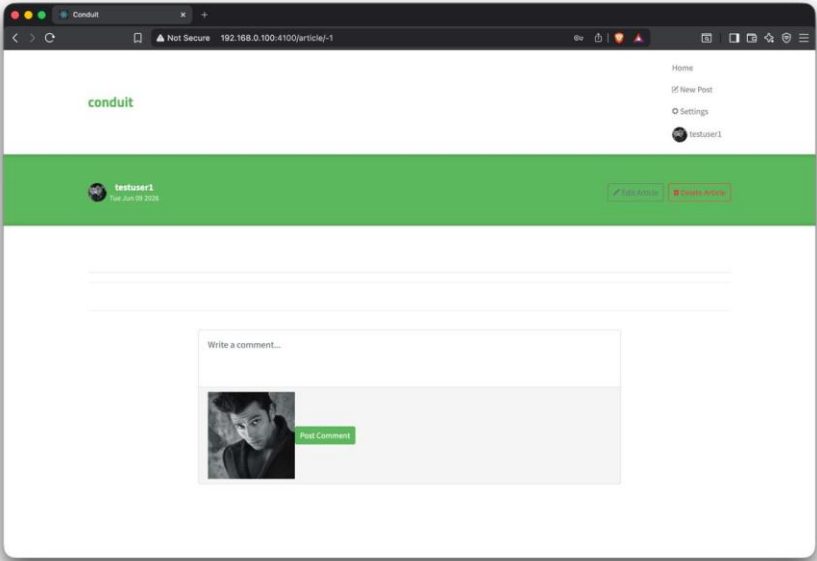
Charter ID	CH-003
Tester Name	Nai Panhavorn
Mission	Test concurrent and boundary behaviors using two browsers logged in to the same account
Areas to Explore	Two browsers same account, rapid clicking, follow and unfollow quickly, edit same article in two tabs
Start Time	01:00 AM
End Time	01:27 AM
Duration	27 minutes
What I Tried	<i>You tested concurrent behavior using two browsers logged into the same account. You performed actions such as opening the same article in both browsers and deleting an article from the first browser while keeping the second browser open. You then refreshed the second browser to check whether the changes were synchronized.</i>
Findings / Bugs	<i>When you edited an article in the first browser, the change was not immediately reflected in the second browser. After refreshing the second browser, an error occurred instead of properly updating the page or removing the deleted article. This indicates a synchronization or state-handling issue between sessions.</i>
Observations	- <i>Changes are not synced between browsers in real time.</i> - <i>After deleting an article in one browser, refreshing the second browser shows an error instead of updating properly.</i>
Screenshots	

Phase 1C — Bug Reports

Fill one table per bug found during testing. Add more tables if needed.

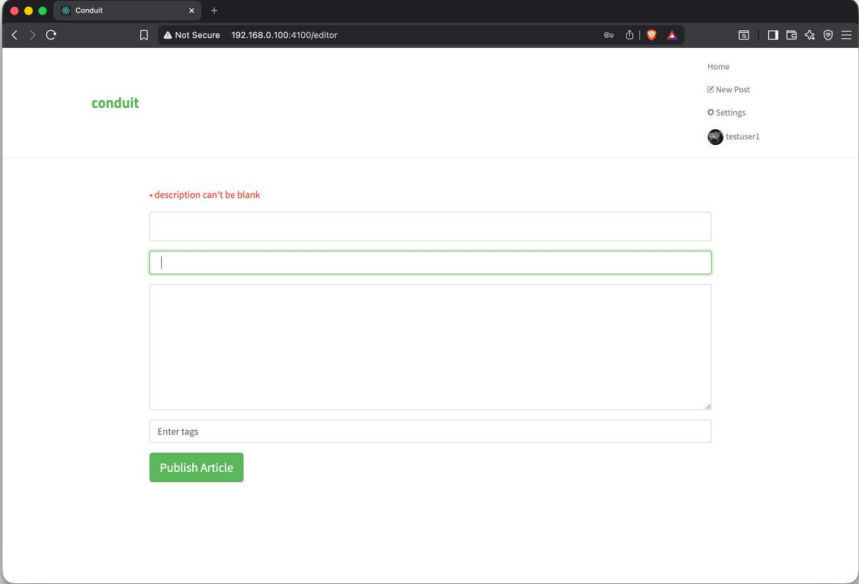
BUG-001

Bug ID	BUG-001
Bug Title	<i>Article Published with No Title or Content</i>
Reported By	<i>Gnan Antopou</i>
Severity	<i>Major</i>
Priority	<i>P2</i>
Environment	Browser: Brave OS: Windows App: localhost:4100
Steps to Reproduce	1. Login to the app 2. Click New Post in navbar 3. Leave all fields completely empty 4. Click Publish Article
Expected Result	App shows an error message. Article is not published.
Actual Result	Article was published successfully with no title and no content. Page shows blank article with URL /article/-1
Evidence	

	
Status	New

BUG-002

Bug ID	BUG-001
Bug Title	<i>Title Field Has No Validation — Only Description Error Shows</i>
Reported By	<i>Gnan Antopou</i>
Severity	Minor
Priority	P3
Environment	Browser: Brave OS: Windows App: localhost:4100
Steps to Reproduce	1. Login to the app 2. Click New Post in navbar 3. Leave title and description both empty 4. Click Publish Article
Expected Result	Error messages appear for both title and description fields
Actual Result	Only "description can't be blank" error appears. No validation error shown for the title field

Evidence	
Status	New

Phase 2 — Engineering

2A —Unit Tests

Developer: Than Soryapongs

File: /unit-tests/users.test.js

Framework: Jest + Supertest

Total: 10 test cases — 6 happy path, 4 sad path

```
macbookpro@Kettos-MacBook-Pro backend % npx jest --runTestsByPath src/tests/users.test.ts
```

```
PASS  api  src/tests/users.test.ts
```

Users and Articles API checklist tests

- ✓ UT-001 Register with valid data (Happy Path) (16 ms)
- ✓ UT-002 Register with duplicate email (Sad Path) (4 ms)
- ✓ UT-003 Login with correct credentials (Happy Path) (2 ms)
- ✓ UT-004 Login with wrong password (Sad Path) (2 ms)
- ✓ UT-005 Login with missing email (Sad Path) (2 ms)
- ✓ UT-006 Create article with valid data (Happy Path) (2 ms)
- ✓ UT-007 Create article without title (Sad Path) (2 ms)
- ✓ UT-008 Create article without auth token (Sad Path) (2 ms)

- ✓ UT-009 Get all articles (Happy Path) (2 ms)
- ✓ UT-010 Delete article with valid auth (Happy Path) (2 ms)

Test Suites: 1 passed, 1 total

Tests: 10 passed, 10 total

Snapshots: 0 total

Time: 0.578 s, estimated 1 s

Ran all test suites within paths "src/tests/users.test.ts".

Test ID	Test Name	Type	Result
UT-001	Register with valid data	Happy Path	Pass
UT-002	Register with duplicate email	Sad Path	Pass
UT-003	Login with correct credentials	Happy Path	Pass
UT-004	Login with wrong password	Sad Path	Pass
UT-005	Login with missing email	Sad Path	Pass
UT-006	Create article with valid data	Happy Path	Pass
UT-007	Create article without title	Sad Path	Pass
UT-008	Create article without auth token	Sad Path	Pass
UT-009	Get all articles	Happy Path	Pass
UT-010	Delete article with valid auth	Happy Path	Pass

2B — Postman API Collection

Developer: Than Soryapongs

File: /postman/Group3_Conduit.postman_collection.json

Base URL: http://localhost:3000/api

Total requests: 10 | Each request has minimum 2 assertions

Search Postman

Invite

Upgrade

POST 1 - POST /users/login

Group3_ConduIt / 1 - POST /users/login

Save

Share

POST

{{baseUrl}}/users/login

Send

</>

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Schema

Beautify

1 {
2 | "user": { "email": "member3@test.com", "password": "Test1234" }
3 }

Body

Cookies

Headers (8)

Test Results (2/2)

200 OK

92 ms

538 B

Save Response

{ } JSON

Preview

Visualize

1 {
2 | "user": {
3 | "email": "member3@test.com",
4 | "username": "member3test",
5 | "bio": null,
6 | "image": "https://i.pravatar.cc/150",
7 | "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyIjpbImkIj0xfSwiaWF0IjoxNzgwNzMyYzYwLmJleHA1OjE3ODU5MTY3NjB9.
8 | ep86Gc0qsW0DYx0skpL2fwaDf27SETU21YN8HJr7Pko"
9 | }

nal

Import Complete

Runner

Start Proxy

Cookies

Vault

Trash

Group3_Conduit / 2 - POST /users

POST{{baseUri}}/users

Send

DocsParamsAuthorizationHeaders (10)BodyScriptsSettings

noneform-datax-www-form-urlencodedrawbinaryGraphQLJSON

SchemaBeautify

```
1 {
2   "user": { "username": "testuser1", "email": "testuser1@example.com", "password": "Password123!" }
3 }
```

BodyCookiesHeaders (8)Test Results (2/2)

201 Created · 96 ms · 553 B

Save Response

JSONPreviewVisualize

```
1 {
2   "user": {
3     "id": 3,
4     "email": "testuser1@example.com",
5     "username": "testuser1",
6     "bio": null,
7     "image": "https://i.pravatar.cc/150",
8     "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyIjpbImkIjozF5wIWF0IjoxNzg4MzMyOTQ1LCJleHAiOjE3ODU5MTY5NDV9.BHquZjs0NyWk1HfG0yZ2BBA31SMJK_LFRF4CYxMBoU"
9   }
10 }
```

Import Complete

RunnerStart ProxyCookiesVaultTrash

Group3_Conduit / 3 - GET /user

GET{{baseUri}}/user

Send

DocsParamsAuthorizationHeaders (8)BodyScriptsSettings

Headers7 hidden

Key	Value	Description
☒ Authorization	Token {{token}}	
Key	Value	Description

BodyCookiesHeaders (8)Test Results (1/2)

200 OK · 14 ms · 545 B

Save Response

JSONPreviewVisualize

```
1 {
2   "user": {
3     "id": 1,
4     "email": "member3@test.com",
5     "username": "member3test",
6     "bio": null,
7     "image": "https://i.pravatar.cc/150",
8     "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyIjpbImkIjozF5wIWF0IjoxNzg4MzMyOTQ1LCJleHAiOjE3ODU5MTY5NDV9.BHquZjs0NyWk1HfG0yZ2BBA31SMJK_LFRF4CYxMBoU"
9   }
10 }
```

Import Complete

RunnerStart ProxyCookiesVaultTrash

Group3_Conduit / 4 - PUT /user

PUT{{baseUrl}}/user

Send

DocsParamsAuthorizationHeaders (11)BodyScriptsSettings

Headers9 hidden

Key	Value	Description
☒ Content-Type	application/json	
☒ Authorization	Token {{token}}	
Key	Value	Description

BodyCookiesHeaders (8)Test Results (2/2)

200 OK17 ms554 B

Save Response

JSONPreviewVisualize

```
1 {
2   "user": {
3     "id": 1,
4     "email": "member3@test.com",
5     "username": "member3test",
6     "bio": "Updated bio",
7     "image": "https://i.pravatar.cc/150",
8     "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ3c2VyIjpbImIjoxfSwiaWF0IjoxNzgwMzMyOTk4L3JleHAiOjE3ODU5MTY5OTU5LnQms0-qTYoTlQKtZZNPpHk_bGYNd2VE0a-2v_T-0c"
9   }
10 }
```

Import CompleteRunnerStart ProxyCookiesVaultTrash

Group3_Conduit / 5 - POST /articles

POST{{baseUrl}}/articles

Send

DocsParamsAuthorizationHeaders (11)BodyScriptsSettings

Query Params

Key	Value	Description
Key	Value	Description

BodyCookiesHeaders (8)Test Results (2/2)

201 Created32 ms625 B

Save Response

JSONPreviewVisualize

```
1 {
2   "article": {
3     "slug": "Postman-Article-1",
4     "title": "Postman Article",
5     "description": "desc",
6     "body": "content",
7     "tagList": [
8       "postman"
9     ],
10    "createdAt": "2026-06-06T08:03:39.192Z",
11    "updatedAt": "2026-06-06T08:03:39.192Z",
12    "favorited": false,
13    "favoritesCount": 0,
14    "author": {
15      "username": "member3test",
16      "bio": "Updated bio",
17      "image": "https://i.pravatar.cc/150",
18      "following": false
19    }
20  }
21 }
```

Import CompleteRunnerStart ProxyCookiesVaultTrash

Group3_Conduit / 6 - GET /articles

SaveShare

GET{{baseUrl}}/articlesSend

DocsParamsAuthorizationHeaders (8)BodyScriptsSettings

Cookies

Document this request... Write with AI

BodyCookiesHeaders (8)Test Results (2/2)

200 OK · 19 ms · 300 BSave Response

JSONPreviewVisualize

```
1 {
2   "articles": [],
3   "articlesCount": 0
4 }
```

Import CompleteRunnerStart ProxyCookiesVaultTrash

POST 1 - POST /users/loPOST 2 - POST /usersGET 3 - GET /userPUT 4 - PUT /userPOST 5 - POST /articlesGET 6 - GET /articlesGET 7 - GET /articles/slug+No environment

Group3_Conduit / 7 - GET /articles/slug

SaveShare

GET{{baseUrl}}/articles/{{articleSlug}}Send

DocsParamsAuthorization

Postman-Article-1

Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

BodyCookiesHeaders (8)Test Results (1/2)

200 OK · 23 ms · 620 BSave Response

JSONPreviewVisualize

```
1 {
2   "article": {
3     "slug": "Postman-Article-1",
4     "title": "Postman Article",
5     "description": "desc",
6     "body": "content",
7     "tagList": [
8       "postman"
9     ],
10    "createdAt": "2026-06-06T08:03:39.192Z",
11    "updatedAt": "2026-06-06T08:03:39.192Z",
12    "favorited": false,
13    "favoritesCount": 0,
14    "author": {
15      "username": "member3test",
16      "bio": "Updated bio",
17      "image": "https://i.pravatar.cc/150",
18      "following": false
19    }
20  }
21 }
```

Group3_Conduit / 8 - PUT /articles/slug

PUT Send

Docs Params Authorization

Postman-Article-1

none form-data x-www-form-urlencoded Collection Variables in request →

Schema Beautify

```
1 {
2   "article": { "title": "Updated Title" }
3 }
```

Body Cookies Headers (8) Test Results (2/2)

200 OK · 27 ms · 607 B

Save Response

JSON Preview Visualize

```
1 {
2   "article": {
3     "slug": "Updated-Title-1",
4     "title": "Updated Title",
5     "description": "desc",
6     "body": "content",
7     "tagList": [],
8     "createdAt": "2026-06-06T08:03:39.192Z",
9     "updatedAt": "2026-06-06T08:07:22.437Z",
10    "favorited": false,
11    "favoritesCount": 0,
12    "author": {
13      "username": "member3test",
14      "bio": "Updated bio",
15      "image": "https://i.pravatar.cc/150",
16      "following": false
17    }
18  }
19 }
```

Import Complete

Runner Start Proxy Cookies Vault Trash

Group3_Conduit / 9 - DELETE /articles/slug

DELETE Send

Docs Params Authorization

Updated-Title-1

Collection Variables in request →

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers (6) Test Results (1/1)

204 No Content · 11 ms · 207 B

Save Response

Raw Preview Visualize

1

Import Complete

Runner Start Proxy Cookies Vault Trash

Group3_Conduit / 10 - POST /articles/:slug/comments

POST {{baseUrl}} /articles/ {{articleSlug}} /comments

Send

Docs Params Authorization Headers (11) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

Schema Beautify

```
1 {
2   "comment": { "body": "Nice article!" }
3 }
```

Body Cookies Headers (8) Test Results (2/2)

200 OK · 17 ms · 499 B

Save Response

JSON Preview Visualize

```
1 {
2   "comment": {
3     "id": 3,
4     "createdAt": "2026-06-06T08:10:05.367Z",
5     "updatedAt": "2026-06-06T08:10:05.367Z",
6     "body": "Nice article!",
7     "author": {
8       "username": "member3test",
9       "bio": "Updated bio",
10      "image": "https://i.pravatar.cc/150",
11      "following": false
12    }
13  }
14 }
```

Runner Start Proxy Cookies Vault Trash

#	Method	Endpoint	Assertions	Result
1	POST	/users/login	Status 200, token exists	Pass
2	POST	/users	Status 201, user object	Pass
3	GET	/user	Status 200, email matches	Pass
4	PUT	/user	Status 200, updated field	Pass
5	POST	/articles	Status 201, slug exists	Pass
6	GET	/articles	Status 200, articles array	Pass
7	GET	/articles/:slug	Status 200, correct slug	Pass
8	PUT	/articles/:slug	Status 200, updated title	Pass
9	DELETE	/articles/:slug	Status 204, No Content	Pass
10	POST	/articles/:slug/comments	Status 200, comment body	Pass

2C — Playwright E2E Automation

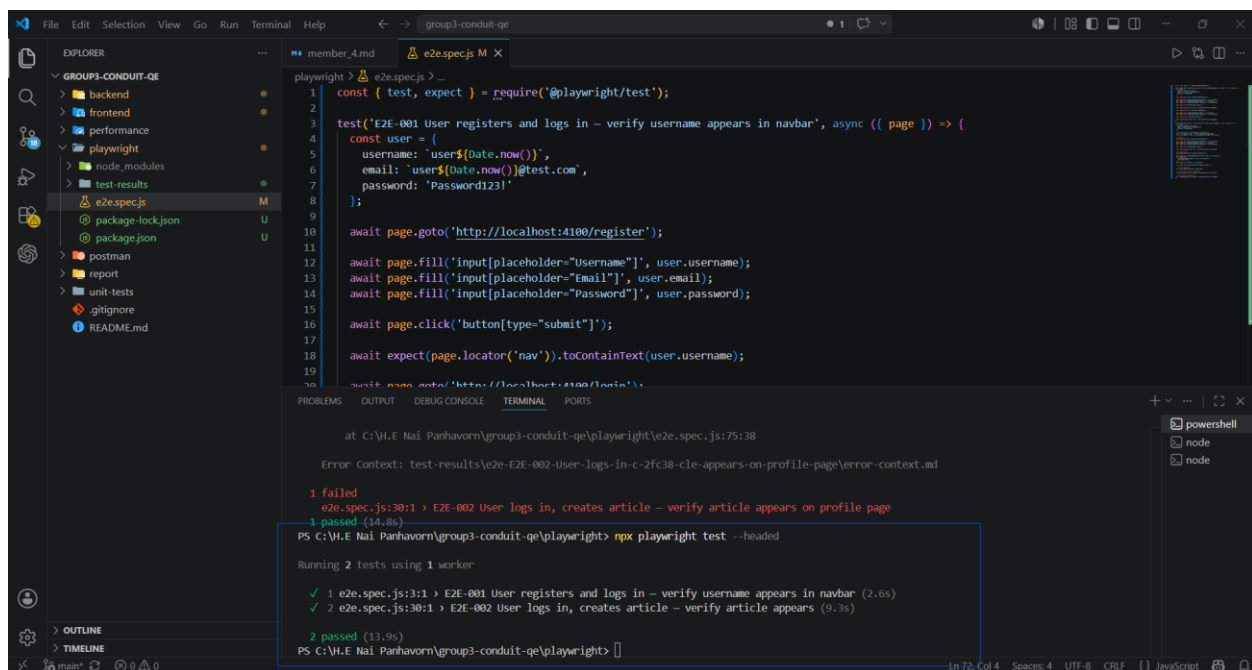
Developer: Nai Panhavorn

File: /playwright/e2e.spec.js

Framework: Playwright

(Paste Playwright test results screenshot below this line)

Flow ID	Description	Result
E2E-001	User registers and logs in — verify username appears in navbar	Pass
E2E-002	User logs in, creates article — verify article appears on profile page	Pass



```
const { test, expect } = require('@playwright/test');

test('E2E-001 User registers and logs in - verify username appears in navbar', async ({ page }) => {
  const user = {
    username: `user${Date.now()}`,
    email: `user${Date.now()}@test.com`,
    password: 'Password123!'
  };
  await page.goto('http://localhost:4100/register');
  await page.fill('input[placeholder="Username"]', user.username);
  await page.fill('input[placeholder="Email"]', user.email);
  await page.fill('input[placeholder="Password"]', user.password);
  await page.click('button[type="submit"]');
  await expect(page.locator('nav')).toContainText(user.username);
  await page.goto('http://localhost:4100/login');
});

test('E2E-002 User logs in, creates article - verify article appears on profile page', async ({ page }) => {
  const user = {
    username: `user${Date.now()}`,
    email: `user${Date.now()}@test.com`,
    password: 'Password123!'
  };
  await page.goto('http://localhost:4100/login');
  await page.fill('input[placeholder="Username"]', user.username);
  await page.fill('input[placeholder="Password"]', user.password);
  await page.click('button[type="submit"]');
  await page.goto('http://localhost:4100/profile');
  await page.click('button[type="button"]');
  await page.fill('input[placeholder="Title"]', 'My Article');
  await page.fill('input[placeholder="Content"]', 'This is my article content');
  await page.click('button[type="button"]');
  await expect(page.locator('h2')).toContainText('My Article');
});
```

```
at C:\V\H\Nai Panhavorn\group3-conduit-qe\playwright\e2e.spec.js:75:38
Error Context: test-results\e2e-E2E-002-User-logs-in-c-2fc38-cle-appears-on-profile-page\error-context.md

1 failed
e2e.spec.js:30:1 > E2E-002 User logs in, creates article - verify article appears on profile page
1 passed (14.8s)

PS C:\V\H\Nai Panhavorn\group3-conduit-qe\playwright> npx playwright test --headed

Running 2 tests using 1 worker

✓ 1 e2e.spec.js:3:1 > E2E-001 User registers and logs in - verify username appears in navbar (2.6s)
✓ 2 e2e.spec.js:30:1 > E2E-002 User logs in, creates article - verify article appears (9.3s)

2 passed (13.9s)

PS C:\V\H\Nai Panhavorn\group3-conduit-qe\playwright>
```

Phase 3 — Performance Analysis

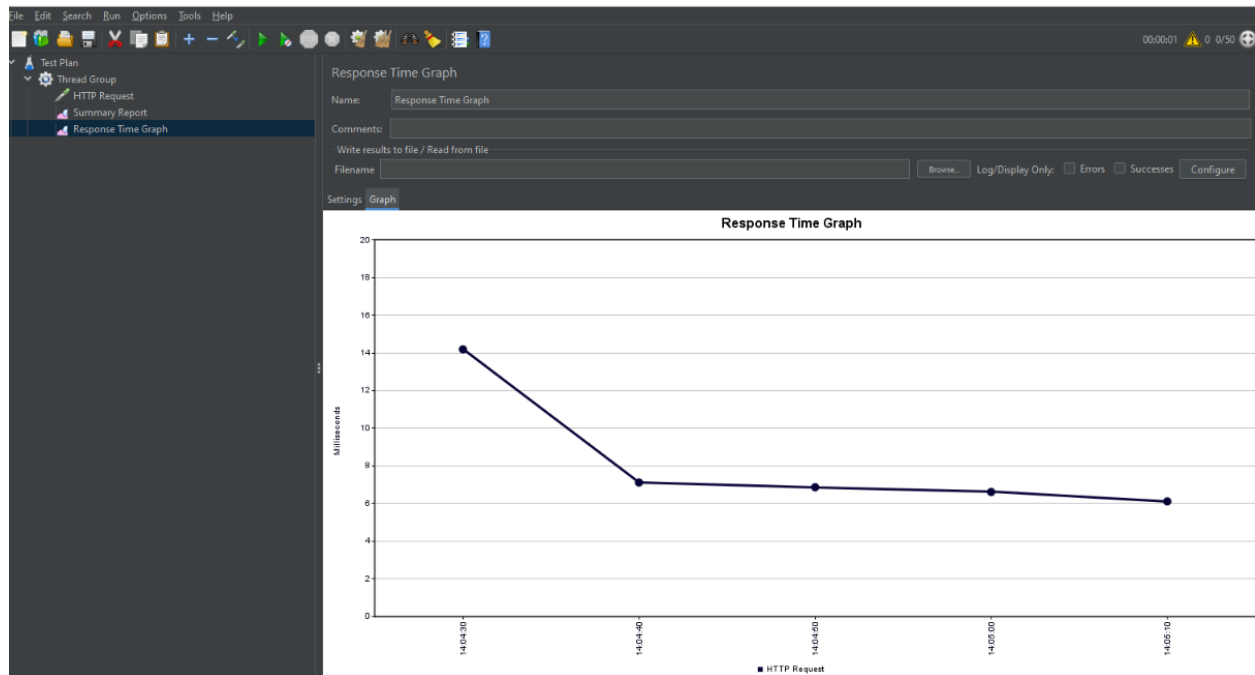
Tester: *Hor Bunnareth*

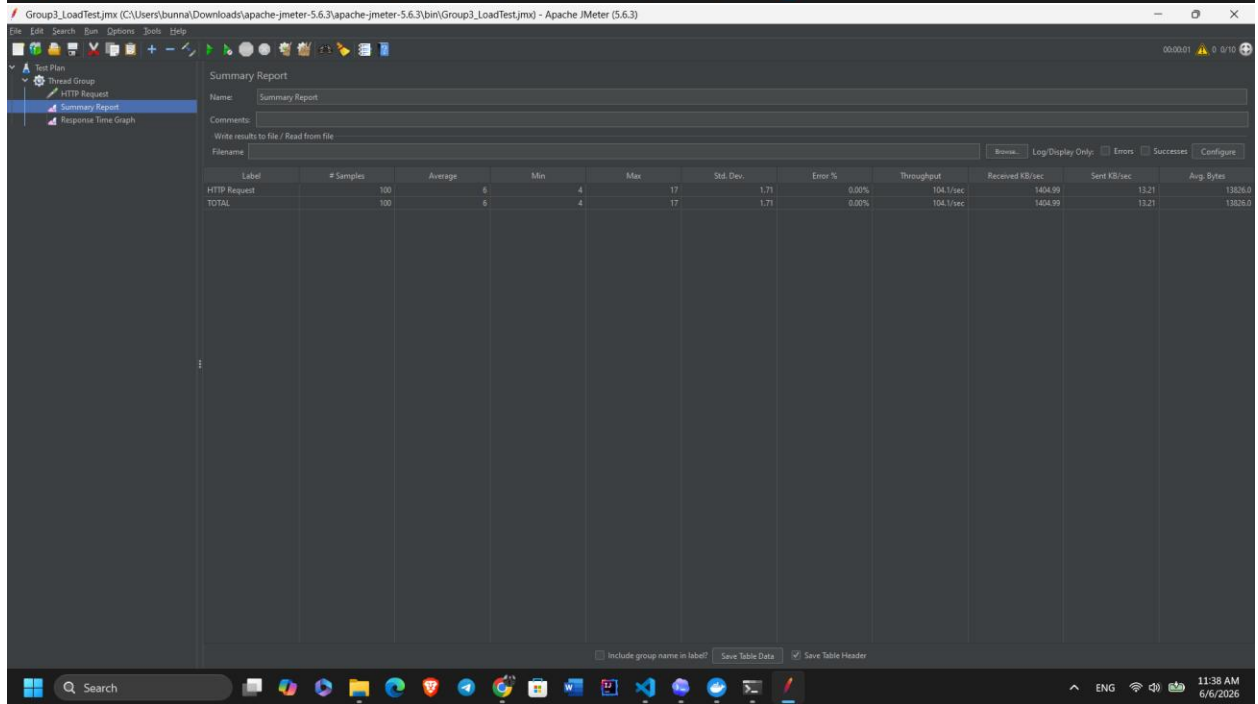
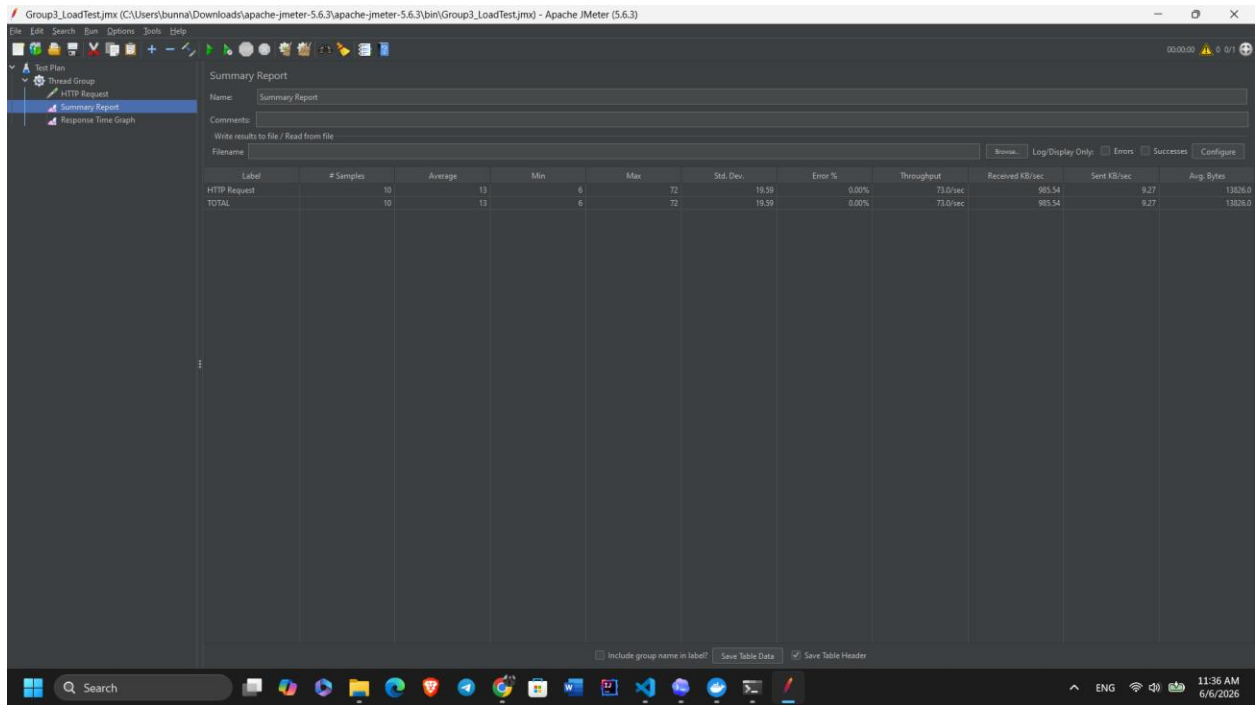
Tool: JMeter

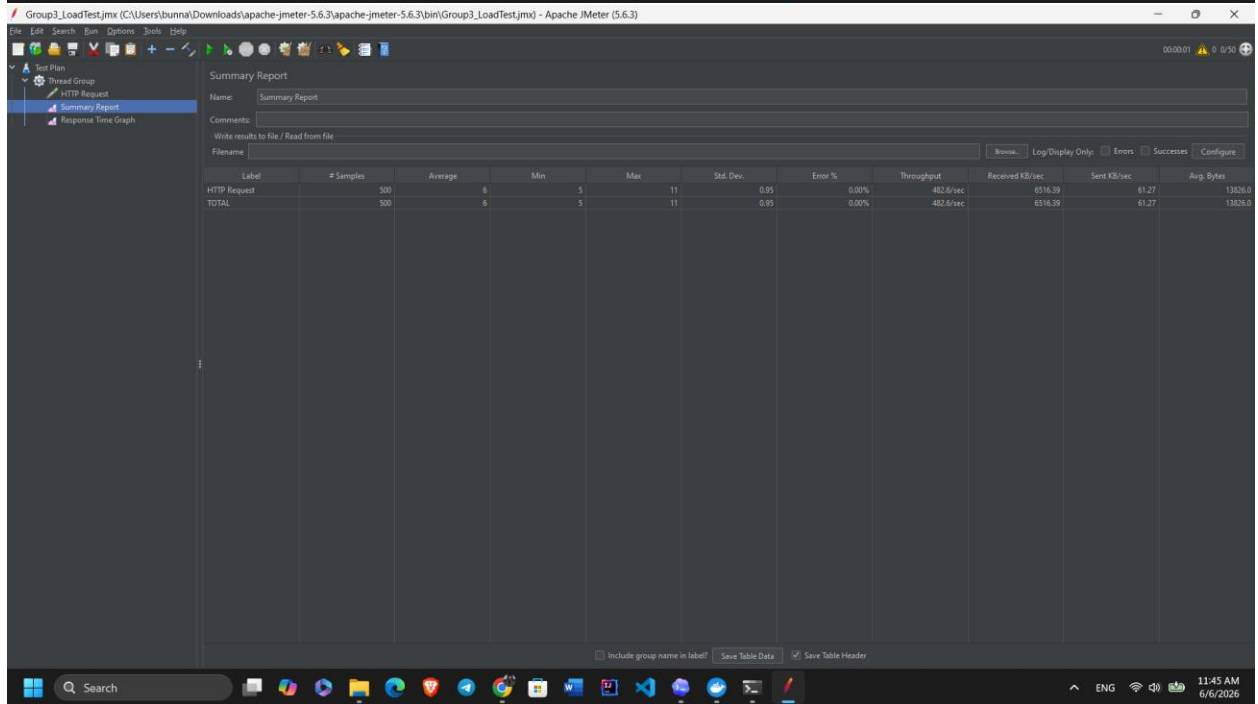
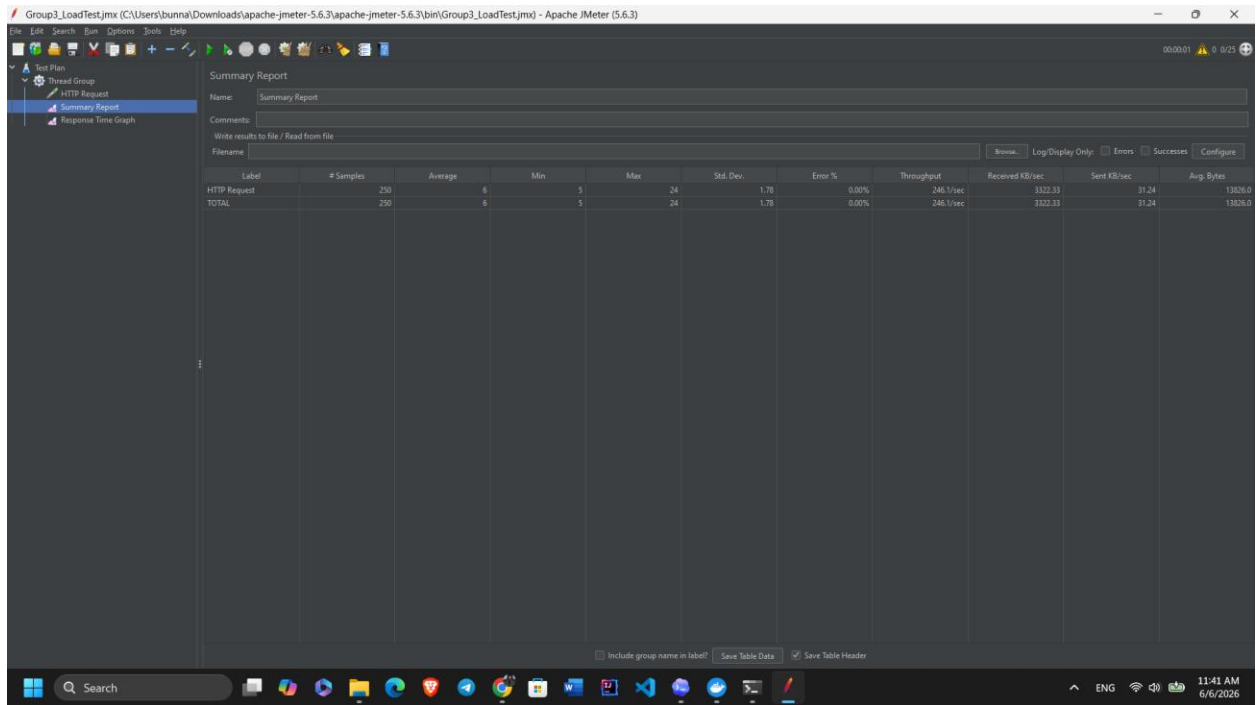
Target: GET <http://localhost:3000/api/articles>

Test Round	Concurrent Users	Avg Response Time	Error Rate
Baseline	1 user	(13 ms)	(0.00%)
Load Test 1	10 users	(6ms)	(0.00%)
Load Test 2	25 users	(6 ms)	(0.00%)
Load Test 3	50 users	(6 ms)	(0.00%)

Response Time vs Load Graph







Bottleneck Analysis

Breaking point (number of users): (Not reached during the test, up to 50 concurrent users.)

Primary bottleneck: (No bottleneck was observed during the tested load range.)

Summary:

The application did not show a slowdown during the tested load range from 1 to 50 concurrent users. The average response time stayed low, around 6-13 ms, and the error rate remained 0.00% for all rounds. Based on these results, the system did not reach its breaking point during this JMeter test. A higher number of users or a longer test duration would be needed to identify the actual bottleneck.

GitHub Repository

Repository link: <https://github.com/Antopou/group3-conduit-qe>

Folder / File	Description
/backend/	Conduit Node.js + Express source code committed by Antopou
/frontend/	Conduit React.js source code committed by Antopou
/unit-tests/users.test.js	Jest unit tests committed by Soryapongs
/postman/*.json	Postman collection export committed by Soryapongs
/playwright/e2e.spec.js	Playwright E2E scripts committed by Panhavorn
/performance/*.jmx	JMeter test plan committed by Bunnareth
/report/*.docx	This report each member commits their section